Q1.

This question is about the periodic table.

In 1864 John Newlands suggested an arrangement of elements.

Figure 1 shows the arrangement Newlands suggested.

Figure 1

1	2	3	4	5	6	7
H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca				

(a) Give **two** differences between column 1 in **Figure 1** and Group 1 in the modern periodic table.

Use the periodic table to help you.

1. _____

_____

2. _____

_____

(2)

(b) In 1869 Mendeleev produced his periodic table.

Complete the sentence.

Choose the answer from the box.

insoluble	**magnetic**	**undiscovered**	**unreactive**

Mendeleev left gaps in his periodic table for elements that were

_____ .

(1)

(c) How are the elements ordered in the modern periodic table?

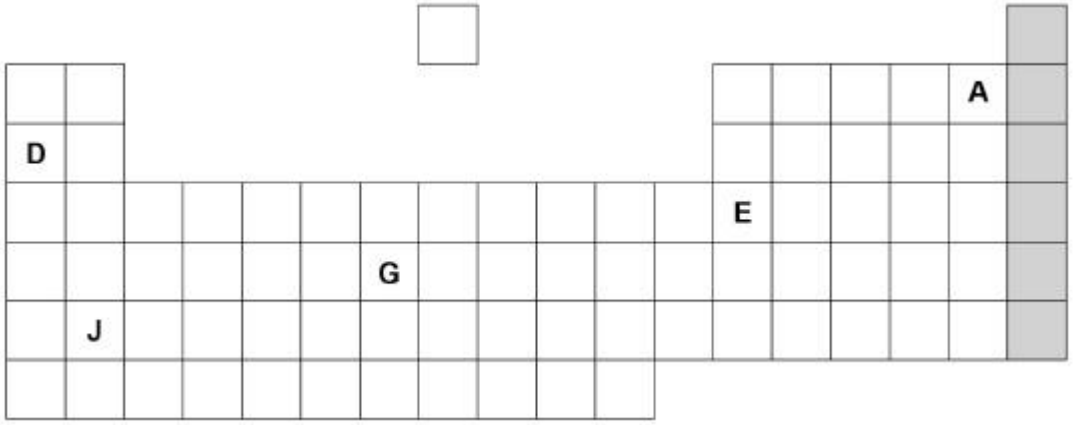
Tick **one** box.

- Atomic mass ☐
- Atomic number ☐
- Melting point ☐
- Reactivity ☐

(1)

Figure 2 shows part of the modern periodic table.

Figure 2



(d) Complete the sentences about the elements in **Figure 2**.

Choose the answers from the box.

A	**D**	**E**	**G**	**J**

Sodium is an alkali metal and is represented by the letter _____.

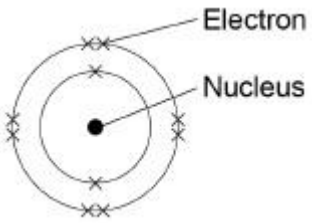
An element in group 3 is represented by the letter _____.

A gaseous non-metal element is represented by the letter _____.

(3)

(e) **Figure 3** shows the electronic structure of an atom.

Figure 3



This element is in the shaded group on **Figure 2**.

Why is this element unreactive?

_____

_____

(1)

(f) Name the group of elements in the shaded column on **Figure 2**.

_____

(1)

(Total 9 marks)

Q1.

(a) any **two** from:

- hydrogen is in group 1 on Newlands table
- fluorine / chlorine / halogens are in group 1 on Newlands table
- alkali metals are in group 2 on Newlands table

allow converse arguments relating to modern table

allow lithium / sodium / potassium for alkali metals

2

(b) undiscovered

1

(c) atomic number

1

(d) **D**

1

E

1

A

1

must be in this order

(e) has a complete outer shell of electrons

allow because has a stable arrangement of electrons

1

(f) noble gases

1

[9]